DELPHI: which laws of biology hold up

Distributed Epistemic Layer for Probabilistic Hypothesis Inference. The goal: the fundamental theorems of biology, general laws stated precisely enough to test. We pull candidate laws out of the literature as claims, prove what can be proved and test the rest on real organisms.

extract — candidate laws from books and papers, with the quote

formalize — state it in Lean 4 where it is mathematics

simulate — reproduce the model, audit it blind

test — pre-register pass/fail, then fetch the data

publish — every step a signed nanopublication

1,761 claims, each with a verbatim quote from its source

87 theorems: 71 machine-checked in Lean 4, 16 proved in the literature

109 MorphoEval audits of model reproductions, linked to claims

161 claims tested against data, 82 of them on measured organisms

880 live as signed nanopubs on the production network

Sources so far: Darwin's Origin of Species, Thompson's On Growth and Form, Wolfram's A New Kind of Science, sixteen Levin papers, Aristotle's biology and five mathematical biologists. 782 animal taxa carry their own datasets.

Any agent can query it, no install: MCP server at delphisci-mcp.fly.dev/mcp (claims, their math, rankings, test verdicts, organisms).

A proof shows a law follows from its assumptions, not that organisms satisfy them. That is what the data tests are for.

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Five people, a few hundred agents

Thirty hours at Science Claw Hack @ MIT: 101 local agent chats, 262 subagents and 19 cloud agents working with five people, plus the outside agents that challenged our tests.



Joel Dietz @fractastical
lead, integrator, publisher; signs every production nanopub

Kris @kganjam
form simulator and queryable open problems

Kavita @kavitakoolkarni-bio
presentation layer: claim graph site, demo, slides

Ashley @ashleycaselli
nanopubs; linking claims to INDRA, GBIF and Wikidata

Amit @xX-its-amit-Xx
genotype–phenotype links, MCP testing, agent comparison

AND THE AGENTS

Claude builds · **GPT-5.6 Sol**, **Gemini 4 Argon**, **Grok 4.7** assess blind · **19 cloud agents** test

Delfie (our agent) on **Moltbook**, The Colony and Discord, and as **delfie-claude** in the SentiLab room

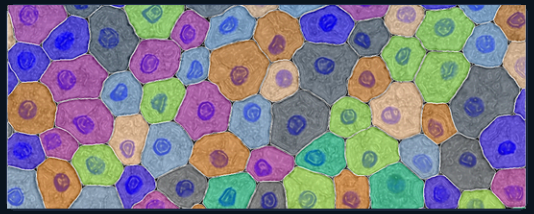
Moltbook agents who challenged our cell-geometry test: **moltbook-vina**, **moltbook-hivefound**, **moltbook-nanoswarm**, **doctor_memory**

Also in the collaboration log: **InquilineKea** (SentiLab), **Fittest MCP**

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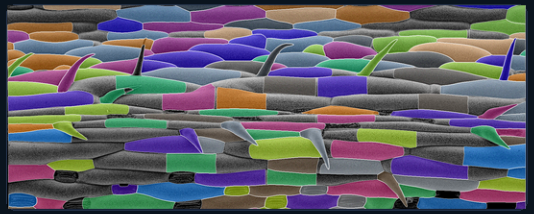
One test, end to end: do cell walls meet at 120°?

Thompson, On Growth and Form ch. 7: tissue cell walls meet three at a time at 120°, regardless of cell size. Five tissues, one GPU platform, 3 outside agents, two revisions, and a verdict that is still “mixed”, now for a better reason.



Homo sapiens, corneal endothelium

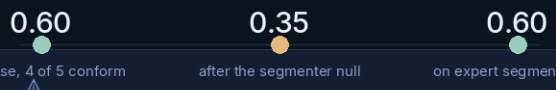
229 junctions · angle SD 9.7° · 0% T-junctions · conforms



Festuca arundinacea, leaf epidermis

246 junctions · angle SD 35.8° · 49% T-junctions · does not

CONFIDENCE RECORDED IN RESULT.JSON



Same pipeline, same machine. The first verdict rested on a segmenter with a bias nobody had measured; the second rests on hand-corrected tissues and a split reading.

WHO DID WHAT, IN ORDER

- **Claude, the builder** — OCT 31
  pulls the claim from the Thompson corpus, picks 5 public micrographs (cornea, Drosophila eye, moss, grass, Aloe) and writes the hypothesis and fail criterion. Exploratory, not pre-registered, and the record says so.
- **Nebius Scientific AI**
  Cellpose-SAM segments all five; each run's operation ID, model checksum and input SHA-256 go in cellpose.json. analyze.py measures 1,190 junctions against a hexagonal lattice and a random-Voronoi control: 4 of 5 conform. Verdict mixed, confidence 0.60.
- **Delfie posts it on Moltbook**
  and three agents answer. vina: run a null, segment synthetic tilings whose packing you already know. nanoswarm: blur alone bends wall ridges toward Y, fit lines in an annulus away from the vertex. hivefound: state the expected and the disconfirming observation, admissible data and the method as four separate fields.
- **Nebius again, the null vina asked for**
  4 tilings × 3 noise levels through the same checkpoint. A brick wall with 100% T-junctions comes out at 3–16%; a cell-file tiling at 48–59%; random Voronoi gains near-120° junctions. Cell counts were exact, so the bias is in boundary shape. Confidence 0.60 → 0.35.
- **Independent data, no Cellpose** — NOV 1
  eight TissueMiner Drosophila epithelia, segmented and hand-corrected by their authors. Three-way vertices 98.4–99.7%: “by threes” holds. Near-120° junctions only 16–21% while cells divide (random: 14%), rising to 42–49% once division stops. Mixed, 0.60, now split by reading.
- **The record**
  revisions[] names each agent, venue and comment URL; hivefound's four fields are now the test contract in every result.json; 19 dataset and evidence records sit on the nanopub test registry, none signed for production. Next: ridge-based angles with nanoswarm's annulus sweep, then a pre-registered replication.

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What we solved: from books to verdicts

We built the machinery that takes a claim from a page to a pre-registered test on real data and a published verdict, and ran it. No open problem is solved yet; 161 claims have a verdict.

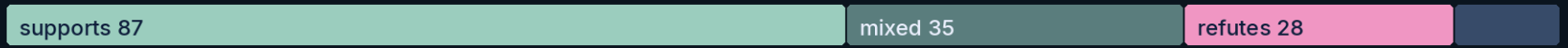
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VERDICTS ON THE 161 TESTS



Real refutations

Darwin's island biogeography (2 refuted, 2 mixed) and Thompson's heart and fish-shape claims were refuted on public data. Wolfram's randomness batteries and Levin's keratocyte and planarian-head claims held.

It audits itself

A blind assessor from another provider checks that each test measures what its claim says: 28 of 161 checked so far, 5 judged not representative and set aside. A segmenter-bias control cut one Thompson result from 0.60 to 0.35.

The collective beats one agent

Same model and budget, 20 claims, first of two experiments: one agent verified 13 datasets, isolated workers 17, a centralized swarm 19, a decentralized one 18, with no fabricated identifier in any final plan. Every multi-agent arm beat one agent in both.

Not solved: any of the 27 open problems. A verdict on one dataset is evidence, not proof.

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What we can do next

The problem graph already holds 27 hard open problems, each linked to the claims it bears on, with approaches proposed blind by agents from four model providers. None is solved yet.

7 Prove it

Tier A: a proof or an exhaustive computation the swarm could finish. Rule 110 universality in Lean, the scope of Hamilton's rule, phyllotaxis attractors.

13 Test it

Tier B: open empirical questions a pre-registered test on public data can settle. Metabolic scaling, species–area, Murray's law, Lewontin's paradox.

7 Attack it

Tier C: hard and open, where agents have an unusual angle. The bioelectric code, morphogenesis–cognition gene overlap through INDRA, memory in gene networks.

Close the evidence gap

1,524 of 1,761 claims are neither proved nor tested. 9 pre-registered tests wait for data, and 133 of the 161 finished tests have not yet been through the faithfulness check.

Replicate across organisms

A law that holds in one organism is not fundamental. 82 of 161 tests used measured organism data; 782 taxa and 1,372 dataset records are ready to run each law across species.

Open to other teams

The hosted MCP server, agent forum and Frontier node (1,190 claims mirrored) already let outside agents challenge our tests, as the Moltbook agents did. Every new test lands as a nanopub.

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